

# PRITISH SAHA

☎ +91 6290609505 ✉ [pritish171@gmail.com](mailto:pritish171@gmail.com) in [LinkedIn](#) @ [GitHub](#) 🎓 [Scholar](#) 🌐 [Website](#)

## Education

Indian Institute of Technology, Kharagpur

Kharagpur, West Bengal

B.Tech. in Manufacturing Science & Engineering + M.Tech. in Industrial & Systems Engineering

Expected completion: May'27

**Relevant Courses:** Safety Fundamentals of Generative AI | Operations Research | Probability & Statistics | Linear Algebra

## Research & Engineering Experience

**Research Fellow (MARS 4.0)** | Cambridge AI Safety Hub | UK (Hybrid) | [\[Report\]](#)

Dec'25 – Present

With Prof. Fernando Rosas, studying Bayesian beliefs in transformers and extending experiments to reinforcement learning

- Decoded beliefs from four-layer transformers in fully observed pilots: probe  $R^2 = 0.985\text{--}0.997$  after 100k training updates
- Built **exact Bayesian filters** to track hidden-state uncertainty in input-output streams under full or partial observation
- Implemented **exact POMDP solvers** and reward-switch experiments, with 115 tests for filtering, geometry and control

**Data Science Intern** | Axis Bank, Business Intelligence Unit | Mumbai | [\[Report\]](#)

May'26 – Jul'26

Built a PySpark personal-loan fraud pipeline over 224.8M accounts and 1.31B transfer edges, implementing weighted PageRank

- Pruned the working graph  $\sim 20\times$  with time-respecting 3-hop BFS, retaining connections for  $\sim 80\%$  of loan applicants
- Delivered an explainable four-signal fraud review queue: **5.49 $\times$  mean lift** in the **top 1%** across three monthly cohorts

**Research Intern** | RAAPID INC | Remote | [\[Research\]](#) [\[Patent\]](#) [\[LoR\]](#)

Mar'25 – Jul'26

With Prof. Amitava Das, developed parameter-efficient LLM fine-tuning methods and GPU kernels for adapter optimization

- Prototyped **Triton kernels** for covariance estimates, matrix inversion and gradient preconditioning in LoRA fine-tuning
- Integrated **CUDA streams** and events to schedule covariance inversions and Fisher eigensolves, with PyTorch fallbacks

**Research Intern** | Complex Networks Research Group, IIT Kharagpur | [\[Report\]](#) [\[LoR\]](#)

Jul'25 – May'26

With Prof. Pawan Goyal, studied latent alignment and reward shaping for mathematical reasoning during LLM post-training

- Added **latent alignment losses** to GRPO for Qwen2.5-Math-7B, using LoRA with 16 rollouts per prompt on one H100
- A selected teacher-augmented pilot averaged **79.57%** MATH-500 accuracy (8 samples/problem; **+4.70 pp** vs GRPO)
- A separate post-hoc audit found shaped credit aligned with GRPO (cosine **0.979**); no shaping signal on all-wrong groups

## Publications & Preprints

**Mixture of Chapters** | Accepted, ICLR 2026 NFAM Workshop | [\[Code\]](#) | [\[Paper\]](#)

Mar'26

Tasmay Pankaj Tibrewal, **Pritish Saha**, Ankit Meda, Kunal Singh, Pradeep Moturi

- Co-developed sparse learnable transformer memory; pretrained **150M–1B-parameter models** on up to **15B tokens**
- Limited ARC-C accuracy loss after instruction tuning to **2.68 pp** versus **6.69 pp** for a compute-matched dense baseline

**GRIT: Geometry-aware PEFT** | arXiv Preprint | [\[Code\]](#) | [\[Paper\]](#)

Jan'26

Pritish Saha, C. Rajbangshi, R. Goyal, M. Goyal, A. Deo, B. Roy, N. D. Singh, R. Goswami, A. Das

- Designed a LoRA method with **rank-space preconditioning**, Fisher-guided adapter reprojection and adaptive ranks
- Scored **66.19%** on **GSM8K** versus LoRA's 63.38% on Llama-3.1-8B with **27.8% fewer** effective adapter parameters

## Competitions

**3rd Place** | General Championship Data Analytics, IIT Kharagpur | [\[Report\]](#)

Mar'26

Captained Patel Hall's team to build a conversational analytics dashboard for database queries and automated KPI discovery

- Built text-to-SQL with schema retrieval and repair: **90% execution accuracy** on a 50-query BIRD mini-dev sample
- Automated KPI discovery in a nine-stage LangGraph pipeline; generated **874K synthetic rows** across 58 client profiles

**4th Place** | ISRO GeoNLI, Inter IIT Tech Meet 14.0 | [\[Report\]](#)

Oct'25 – Dec'25

Co-designed a remote-sensing assistant for captioning, grounding and question answering across RGB, SAR and IR imagery

- Designed a Qwen3-VL-8B curriculum with **LoRA SFT** followed by **DPO** for satellite captioning and question answering
- Paired SAM3 with pyramidal tiling and VLM refinement to count objects, estimate areas and answer spatial queries

**National Finalist** | American Express Campus Challenge, Decision Science | [\[Report\]](#)

Jul'25

Built a personalized offer-ranking system from customer histories, combining gradient-boosted trees with a residual transformer

- Scored **0.59 MAP@7** on final evaluation; trained a transformer with listwise loss to correct boosted-tree ranking errors
- Generated **3K+ customer-history features** with point-in-time joins and customer-grouped validation to limit leakage

## Technical Skills

**Languages & Systems:** Python, C++, SQL, CUDA, Triton, PySpark, Hadoop, HDFS, Linux, Docker, SLURM, Git

**ML & Post-training:** PyTorch, JAX, Transformers, LoRA/QLoRA, DPO, TRL, vLLM, FSDP/DTensor, FlashAttention-2

## Extra-Curricular Activities

Karate black belt and tournament winner | Represented Patel Hall in interhall football and water polo | NSS village volunteer